Apache Spark on Databricks

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Big Data Engineering, Spring 2025
Assignment 2

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Executive Summary

This project examines New York City taxi trip records to develop a robust predictive framework for estimating trip fares. Using Databricks on the Azure cloud environment, the analysis involved schema harmonisation, data cleaning, enrichment, descriptive analytics, and machine learning. The findings highlight the importance of systematic preprocessing and careful model evaluation in **large-scale data engineering tasks**.

Key outcomes are summarized below:

- **Data engineering:** A Snowflake-style lakehouse was established, with schema harmonisation across yellow and green taxis, feature engineering (trip duration, distance, and speed), and validation against TLC dictionaries.
- **Business insights:** Monthly patterns, borough-to-borough flows, and tipping behaviours were analysed, providing insight into demand and driver incentives (*see Appendix A for supporting visualisations*).
- **Predictive modelling:** Four models were evaluated—baseline hierarchical averages, Ridge regression (closed-form), Ridge regression (gradient descent), Huber regression, and a decision tree fallback. The Ridge closed-form model outperformed others, yielding the lowest RMSE on validation and test sets.

This work demonstrates both technical competence in PySpark-based analytics and applied skills in model development, evaluation, and interpretability (*see Appendix E for coefficient tables and multicollinearity checks*).

Introduction

The exponential growth of ride-hailing and taxi data has created both opportunities and challenges for urban mobility analytics. Public datasets, such as the **NYC Taxi and Limousine Commission (TLC) trip records**, provide valuable insights into transportation demand, pricing mechanisms, and passenger behaviour (TLC, 2025). However, the raw data is noisy and requires extensive cleaning and enrichment before any predictive or business value can be extracted.

This project was conducted in three phases:

- **Data engineering:** ingestion, harmonisation, and cleaning of TLC records, ensuring compliance with published data dictionaries (TLC, 2025).
- **Business intelligence:** descriptive statistics on trip patterns, seasonal trends, and revenue hotspots to address the guiding questions.
- **Predictive modelling:** development and evaluation of machine learning models to forecast `total_amount`, incorporating both linear and tree-based methods.

By combining these phases, the project aimed to build a reproducible workflow for handling large-scale transportation data while addressing both descriptive and predictive objectives.

Part 1 - Data Ingestion & Preparation

1. Computing Environment and Storage

To ensure reproducibility and operational robustness, we standardised the platform, storage layout, and time configuration as follows. We used the following settings and artefacts:

- **Platform:** Databricks Serverless (general compute, 32 GB RAM) *(Appendix A1)*.
- **Storage:** Unity Catalog **Volumes** under `workspace.bde.assignment2`.
- **Time zone:** Spark session set to **`America/New_York`** so month/day-of-week/hour reflect NYC local time.
- **Key artefacts and locations:**
    - Source parquet files in Volumes: `/Volumes/workspace/bde/assignment2/green/green_taxi.parquet`, `/Volumes/workspace/bde/assignment2/yellow/yellow_taxi.parquet`.
    - Taxi Zone lookup table: `workspace.bde.taxi_zone_lookup`.
    - Final analysis table: `workspace.bde.taxi_trips_cleaned_borough` (Delta).

2. Data Sources and Baseline Size

Before any transformation, we quantified the input data to establish a governance baseline. Baseline row counts were as follows *(Appendix A2)*:

- **Green taxi:** 83,484,688
- **Yellow taxi:** 907,982,776
- **Total: 991,467,464**

3. Data Cleaning

Harmonisation and Feature Engineering

Because yellow and green fleets use distinct field names and contain colour-specific attributes, we standardised both datasets to a common schema and created computed features for quality checks. Schema harmonisation proceeded along these dimensions:

- **Timestamps:** `pickup_datetime`, `dropoff_datetime` (cast to TIMESTAMP).
- **Location keys:** `PULocationID`, `DOLocationID`.
- **Coded fields:** `passenger_count`, `RatecodeID`, `payment_type` cast to **INT**; (green-only) `trip_type` to **INT**.
- **Monetary fields:** retained as **DOUBLE**.
- **Fleet discriminator:** `color ∈ {yellow, green}`.
- **Union policy:** `unionByName(allowMissingColumns=true)` to preserve colour-specific fields (e.g., `airport_fee`, `ehail_fee`, `trip_type`) as NULL where absent.
- **Integrity check:** the unified raw count remained **991,467,464**, matching the baseline.

To support plausibility screening and later KPIs, we derived the following features:

- **`duration_min`** = (dropoff – pickup) in minutes;
- **`trip_distance_km`** = `trip_distance` (miles) × 1.60934;
- **`speed_kmh`** = `trip_distance_km` / duration (hours), safely guarded against zero/negative denominators.

Cleaning Protocol (TLC-Informed; ≤10% Removal)

The cleaning strategy is conservative and guided by TLC documentation, prioritising the removal of physically impossible or corrupted records while retaining realistic edge cases. The rules applied are summarised below:

- **Chronology and plausibility:**
    - `dropoff_ts > pickup_ts`;
    - `1 ≤ duration_min ≤ 180`;
    - `0.1 ≤ trip_distance_km ≤ 200`;
    - `0 < speed_kmh ≤ 120`.
- **Passenger count:** retain when **NULL** or **0–6** (we do **not** force ≥1 because TLC counts can legitimately be zero or missing).
- **Billing:** `total_amount ≥ 0` (zero allowed).
- **Coverage windows (per TLC timeline):**
    - **Yellow:** pickup ≥ **2009-01-01**;
    - **Green:** pickup ≥ **2013-08-01**;
    - For both: dropoff < **2025-01-01** (keeps all 2024, excludes future-dated anomalies).
- **Code domains (data dictionaries):**
    - `RatecodeID ∈ {1,2,3,4,5,6,99}`, `payment_type ∈ {0,1,2,3,4,5,6}`, `trip_type ∈ {1,2}, store_and_fwd_flag ∈ {Y,N}`;
    - **Nulls are retained** for these fields; early years legitimately lack some surcharges/flags.
- **Surcharge timelines (interpretive note, not a drop rule):** improvement surcharge (2015 onward) and congestion surcharge (2019 onward) can be NULL in prior years and are expected.

Removal Audit and Diagnostics

We report the overall reduction and the rule-level diagnostics to evidence that removals are focused on plausibility issues rather than indiscriminate dropping. The overall audit is:

- **raw_rows:** 991,467,464
- **clean_rows: 974,672,551**
- **removed_rows: 16,794,913**
- **removed_pct: 1.69%** *(Appendix A3)*

Interpretation: As expected with meter/clock/mileage artefacts, the duration-, distance-, and speed-based checks dominate. Code-domain violations are negligible after other plausibility filters. The TLC window rule removes a small set of calendar outliers without affecting representativeness.

4. Final Result

Borough Enrichment and Final Persisted Dataset

To enable origin–destination analysis required in Part 2, both ends of each trip were enriched via a LEFT JOIN to the Taxi Zone Lookup. The enrichment and persistence choices are as follows:

- **Join policy:** LEFT JOIN on `PULocationID` and `DOLocationID` with **"Unknown"** assigned where a location ID cannot be mapped (prevents silent loss in OD grids).
- **Physical design:** the final dataset is written as **Delta** and **partitioned by (**`year`**,** `month`**)** to accelerate month-level and time-of-day queries.
- **Final row count: 974,672,551** in `workspace.bde.taxi_trips_cleaned_borough`. *(Appendix A4)*

Assumptions, Limitations, and Compliance

For transparency, we note several interpretive considerations that inform later analyses. Key assumptions and limitations are:

- **Tip amounts** reflect **card tips only**; **cash tips are not captured**, so tip-incidence estimates are conservative.
- **Passenger count** can be **0 or NULL** for legitimate operational reasons; retaining these avoids bias and respects the ≤10% removal constraint.
- **Null surcharges** in early years are aligned with policy start dates and are not treated as errors.
- **Compliance statement:** the cleaning removed **1.69%** of rows, well within the assignment's **≤10%** cap.

Reproducibility and Operational Notes

To support reruns and auditing, we made the process deterministic and storage-resilient. The following measures support reproducibility:

- Idempotent creation of catalog/schema/tables and deterministic view definitions.
- Storage in **Volumes** to survive cluster idling.
- Logged row counts at each stage (baseline, union, cleaned, final).
- Fixed session time zone (`America/New_York`) to ensure consistent temporal features across sessions.

Part 2 - Business Questions

1. Year–Month summary

We summarized activity at the month level, reporting total trips, the busiest **day of week**, the busiest **hour**, and average **passengers**, **amount per trip**, and **amount per passenger** (the last using a safe divide to exclude 0/NULL passenger rows) *(Appendix B1)*.

Interpretation: These patterns - Friday/Saturday dominance and afternoon peaks - are consistent with commuter and leisure flows. Early-year minima reflect partial availability in historical releases.

2. Descriptive statistics by taxi colour

We computed duration (min), distance (km), and speed (km/h) distributions by colour (avg/median/min/max) *(Appendix B2)*.

- **Green:** duration **13.86** avg (**10.68** med; **1–180** range), distance **4.92** avg (**3.17** med; **0.11–199.94**), speed **20.51** avg (**18.59** med; **0.04–119.99**).
- **Yellow:** duration **14.44** avg (**11.32** med), distance **4.92** avg (**2.77** med), speed **18.91** avg (**16.59** med; **0.04–120**).

Interpretation: Green trips skew to outer boroughs (slightly higher speeds and median distances), while yellow trips are more Manhattan-centric (slower, shorter at the median, but similar mean distance).

3. Colour × PU/DO borough × month × DOW × hour grid

This fine-grained grid reports total trips, average distance, average amount per trip, and total amount. After loading the taxi-zone lookup and rebuilding the enriched table, boroughs resolve correctly *(Appendix B3)*.

Interpretation: In later years/months (not shown here), the grid typically concentrates revenue in Manhattan-centered OD pairs with strong daytime and evening peaks.

4. 2024 Top-10 PU to DO borough pairs and revenue share

We ranked 2024 pairs by **total_amount** and computed each pair's **share of 2024 revenue**. Results show a heavy Manhattan concentration *(Appendix B4)*.

Rank	PU Borough	DO Borough	2024 Total Amount	2024 Share
1	Manhattan	Manhattan	$714.78M	62.45%
2	Queens	Manhattan	$174.14M	15.21%
3	Manhattan	Queens	$74.70M	6.53%

4	Queens	Brooklyn	$38.75M	3.39%
5	Queens	Queens	$33.18M	2.90%
6	Manhattan	Brooklyn	$32.97M	2.88%
7	Queens	Unknown	$14.91M	1.30%
8	Manhattan	EWR	$12.66M	1.11%
9	Brooklyn	Brooklyn	$7.90M	0.69%
10	Brooklyn	Manhattan	$7.80M	0.68%

Interpretation: Intra-Manhattan and cross-borough flows with Manhattan endpoints dominate revenue. Airport-related OD pairs also contribute materially.

5. Tipping incidence and high-tip share

Across all trips, **62.94%** include a tip. Among tipped trips, **0.83%** have tips of **≥ 15** (currency: USD) *(Appendix B5)*.

Important caveat: per TLC, `tip_amount` captures **card tips only**; **cash tips are not recorded**, so these are lower-bound estimates of tipping behaviour.

6. Duration bins: average speed and distance per dollar

We grouped trips into duration bins and computed **average speed** and **km per dollar** *(Appendix B6)*.

Duration	Average speed (km/h)	Average distance per dollar (km/$)
Under 5 mins	19.76	0.16
5–10 mins	17.20	0.21
10–20 mins	17.92	0.26
20–30 mins	21.39	0.31
30–60 mins	25.81	0.37
≥ 60 mins	22.74	0.6

Interpretation: km per $ rises with duration because fixed charges are amortized. However, for driver earnings, the relevant metric is **revenue per minute**; this typically favours **shorter trips** due to flag-drop and minimum-fare effects. (Include the optional revenue-per-minute table if you compute it.)

Part 3 - Machine Learning

1. Data Preprocessing

Problem & Data Split

We model **`total_amount`** as a supervised regression task on the cleaned TLC corpus (~975M trips). To preserve temporal causality we use a **time-based split**:

- **Train:** up to 2024-08-31
- **Validation:** 2024-09-01 to 2024-09-30
- **Test (hold-out):** 2024-10-01 to 2024-12-31

Features & Leakage Control

Predictors available at pickup time only:

- **Continuous:** trip distance (km), duration (min), passenger_count, hour, month, day-of-week.
- **Categorical signals: smoothed target encodings** (m-estimate) for `color`, `pu_borough`, `do_borough` (fit on **train only**, with global-mean fallback).
- **Standardization:** all numeric/TE variables **z-scored on train stats**. Per brief, we **exclude** `fare_amount` and `tolls_amount` and avoid other direct target components.

Heavy-Tail Robustness

Labels are heavy-tailed (test p99 ≈ 105; max ≈ 335,551) *(Appendix C1)*. We **train** on a winsorized target (**cap at train p99.95 = 134.82**) *(Appendix C2)* and **evaluate** on both:

- **TRUE RMSE** (original labels; primary for comparison), and
- **ROBUST RMSE** (diagnostic stability).

2. Models Training

Algorithms Selection

Four supervised regression algorithms were selected:

- **Baseline hierarchical averages**: simple benchmark derived from averages across categorical keys.
- **Ridge regression (closed-form)**: regularised linear model estimated analytically.
- **Ridge regression (gradient descent)**: iterative solver providing a comparison to closed-form estimation.
- **Huber regression**: robust linear model balancing squared and absolute errors.
- **Decision tree (histogram fallback)**: non-linear rule-based model trained using quantile binning.

Results and Evaluation

Model	RMSE (Val, TRUE)	RMSE (Test, TRUE)	RMSE (Val, ROBUST)	RMSE (Test, ROBUST)
Baseline (hierarchical means)	16.948	103.391	—	—
Model A — **Ridge (closed-form)**	13.357	102.847	**12.577**	**12.611**
Model B — **Ridge (GD)**	18.231	103.591	17.468	17.504
Model C — **Huber**	29.012	105.900	28.188	28.188
Model D — **Histogram Fallback**	16.652	103.267	15.279	15.036

Performance was assessed using **Root Mean Squared Error (RMSE)** on both the true labels and the robustly clipped labels *(Appendix D)*. Results demonstrated:

- **Ridge Closed-form** consistently achieved the **lowest validation RMSE** (13.36).
- **Decision Tree Historical Fallback** showed competitive performance (16.65) and provided a useful non-linear benchmark.
- **Gradient descent Ridge** and **Huber regression** underperformed relative to both baseline and closed-form ridge.

Technical Interpretability

Interpretability analyses were conducted on Ridge closed-form coefficients, feature correlations, and multicollinearity diagnostics *(Appendix E)*. Results indicated that:

- **Correlations (train, robust):** `trip_distance_km_z` (0.917), `duration_min_z` (0.828), then TE signals (`pu_te_z` 0.428; `do_te_z` 0.393); time variables and passenger_count≈0.
- **Standardized coefficients (Model A):** `dist` **9.307** > `dur` **4.095** >> `do_te` 0.970 ≈ `color_te` 0.879 > `pc` −0.231 > `pu_te` 0.209 > small time effects (`mo`, `hr`, `dw`).
- **Collinearity (VIF):** inspection **did not indicate pathological multicollinearity**; observed dependencies (e.g., distance–duration) are **mitigated by ridge**.

Model Selection

Based on validation RMSE, interpretability, and robustness, the **Ridge regression (closed-form)** model was selected as the best-performing algorithm. Model artifacts, including weights, z-score statistics, target-encoding maps, and metadata, were stored in Delta format for reproducibility.

Conclusion & Recommendations

A compact, leakage-free linear model with lightly engineered features delivers **material gains** over a strong baseline and **stable generalisation** into Oct–Dec 2024. The **ridge (closed-form)** model achieves the best validation accuracy (**RMSE 13.36 vs 16.95** baseline) and essentially identical **robust** performance between validation and test (**12.58** ≈ **12.61**), indicating the model captures stable structure while the elevated “TRUE” test RMSE (~103) is driven by a small fraction of extreme fares. Feature diagnostics (correlations, standardised coefficients, VIF) align with fare mechanics - **distance and duration** dominate; borough encodings add context; temporal signals are modest; collinearity is manageable under ridge.

Limitations

- **Heavy tails & rare events:** Extreme fares (e.g., long airport or traffic incidents) dominate squared error; we chose not to exclude them at evaluation time.
- **Coarse geography:** Borough-level effects abstract away within-borough heterogeneity; granular zone-pair or route features may add lift.
- **Exogenous drivers:** Weather, disruptions, and special events are excluded; these covariates can materially affect demand and fares.
- **Linear specification:** Ridge restricts interactions and non-linearities; some price behaviour may be piecewise or threshold-based.

Future Work

- **Robust/alternative objectives:** Huber or quantile regression to further dampen outlier influence; small λ grid-search.
- **Feature enrichment:** Peak/off-peak indicators; limited interactions (e.g., distance×season); finer **zone-pair** encodings; optional exogenous data (weather/events).
- **Model class exploration:** Gradient-boosted trees on sampled data for non-linear lift; compare AUC-style calibration on high-fare tail.
- **MLOps:** Monthly retraining, **TE map** refresh, RMSE/robust-RMSE and drift dashboards by **month×borough-pair**, with canary rollout and rollback criteria.

References

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Appendices

Appendix A – Data Ingestion and Cleaning

- **A1.** Cluster Configuration

Serverless Starter Warehouse

Overview | Connection details | Monitoring

Status	Running
Name	Serverless Starter Warehouse (ID: c7ea0ad1cd207a94)
Type	Serverless
Cluster size	2X-Small
Auto stop	After *10* minutes of inactivity
Scaling	Cluster count: Active *1* Min *1* Max *1*
Channel	Current (v 2025.25)
Created by	manhtuan.nguyen@student.uts.edu.au

- **A2.** Baseline Confirming Ingestion

```
green_df: pyspark.sql.connect.dataframe.DataFrame = [VendorID: long, lpep_pickup_datetime: timestamp ... 18 more fields]
yellow_df: pyspark.sql.connect.dataframe.DataFrame = [VendorID: long, tpep_pickup_datetime: timestamp ... 17 more fields]
green_count = 83,484,688
yellow_count = 907,982,776
total_count = 991,467,464
```

- **A3.** Removals Result (≤10% cap)

	raw_rows	clean_rows	removed_rows	removed_pct
1	991467464	974672551	16794913	1.69

- **A4.** DDL of workspace.bde.taxi_trips_cleaned_borough and final row count

	final_rows
1	974672551

Appendix B – Descriptive Analytics Outputs

Screenshots and summary tables from Part 2, including:

- **B1.** Year–Month trip summaries

	ym	total_trips	most_trips_dow	most_trips_hour	avg_passengers	avg_amount_per_trip	avg_amount_per_passenger
1	2009-01-01	268	Thursday	5	1.67	21.61	17.74
2	2011-01-01	3	Monday	18	1	13.8	13.8
3	2013-12-01	150213	Tuesday	20	1.85	15.55	11.68
4	2014-01-01	14411074	Friday	14	1.69	14.36	11.64
5	2014-02-01	13922304	Friday	14	1.68	14.5	11.8
6	2014-03-01	16468962	Saturday	15	1.68	14.65	11.93
7	2014-04-01	15783203	Wednesday	15	1.68	14.95	12.14
8	2014-05-01	16102486	Friday	15	1.68	15.52	12.57
9	2014-06-01	14912181	Friday	15	1.68	15.53	12.59
10	2014-07-01	14256408	Thursday	15	1.68	15.2	12.3
11	2014-08-01	13702424	Friday	15	1.68	16.13	13.15
12	2014-09-01	14547766	Tuesday	15	1.66	15.58	12.69
13	2014-10-01	15713098	Friday	15	1.66	15.58	12.71
14	2014-11-01	14581509	Saturday	14	1.66	15.37	12.5
15	2014-12-01	14707185	Wednesday	14	1.67	15.55	12.64

- **B2.** Descriptive statistics by taxi colour

	color	duration_avg_min	duration_med_min	duration_min_min	duration_max_min	dist_avg_km	dist_med_km	dist_min_km	dist_max_k
1	green	13.86	10.68	1.00	180.00	4.92	3.17	0.11	
2	yellow	14.44	11.32	1.00	180.00	4.92	2.77	0.11	

- **B3.** Borough-to-borough trip grid Color

	color	pu_borough	do_borough	ym	dow_name	hr	total_trips	avg_distance_km	avg_amount_per_trip	total_amount
1	yellow	Brooklyn	Manhattan	2009-01-01	Thursday	1	1	11.27	23.8	23.8
2	yellow	Brooklyn	Manhattan	2009-01-01	Thursday	8	1	15.22	33.38	33.38
3	yellow	Brooklyn	Manhattan	2009-01-01	Thursday	9	1	8.85	29.8	29.8
4	yellow	Brooklyn	Queens	2009-01-01	Thursday	17	1	8.58	21.96	21.96
5	yellow	Manhattan	Bronx	2009-01-01	Thursday	10	1	10.41	26.8	26.8
6	yellow	Manhattan	Bronx	2009-01-01	Thursday	14	1	15.64	34.8	34.8
7	yellow	Manhattan	Brooklyn	2009-01-01	Thursday	6	1	3.28	14.8	14.8
8	yellow	Manhattan	Brooklyn	2009-01-01	Thursday	9	1	10.85	24.8	24.8
9	yellow	Manhattan	Brooklyn	2009-01-01	Thursday	12	1	7.03	23.76	23.76
10	yellow	Manhattan	Brooklyn	2009-01-01	Thursday	16	1	15.06	30.3	30.3
11	yellow	Manhattan	Brooklyn	2009-01-01	Thursday	17	2	10.16	24.78	49.56
12	yellow	Manhattan	EWR	2009-01-01	Thursday	10	1	30.32	80.8	80.8
13	yellow	Manhattan	Manhattan	2009-01-01	Thursday	0	10	2.89	11.23	112.26
14	yellow	Manhattan	Manhattan	2009-01-01	Thursday	1	10	3.24	12.95	129.5
15	yellow	Manhattan	Manhattan	2009-01-01	Thursday	2	10	2.24	10.31	103.1

- **B4.** Top-10 revenue pairs

	pu_borough	do_borough	pair_total_amount_2024	share_pct_2024
1	Manhattan	Manhattan	714781074.9	62.45
2	Queens	Manhattan	174144855.27	15.21
3	Manhattan	Queens	74699660.48	6.53
4	Queens	Brooklyn	38753531.5	3.39
5	Queens	Queens	33184688.13	2.9
6	Manhattan	Brooklyn	32974651.58	2.88
7	Queens	Unknown	14911449.85	1.3
8	Manhattan	EWR	12662829.56	1.11
9	Brooklyn	Brooklyn	7895618.21	0.69
10	Brooklyn	Manhattan	7803213.77	0.68

- **B5.** Tipping behaviour

	pct_with_tips	pct_tips_15_plus_among_tipped
1	62.94	0.83

- **B6.** Duration bin analysis

	duration_bin	avg_speed_kmh	avg_km_per_dollar
1	Under 5 mins	19.76	0.1626
2	5–10 mins	17.2	0.21
3	10–20 mins	17.92	0.2573
4	20–30 mins	21.39	0.3055
5	30–60 mins	25.81	0.3749
6	≥60 mins	22.74	0.6

Appendix C – Distribution & Label Clipping

- **C1.** Distribution sanity checks for `total_amount` (train, validation, test)

```
TRAIN
+-----------------+-----------------+-----+-----+-----+-----------+
|mean_amt         |std_amt          |p50  |p90  |p99  |max_amt    |
+-----------------+-----------------+-----+-----+-----+-----------+
|17.427543521744408|376.6759539388135|12.96|32.16|74.39|1.0000015E7|
+-----------------+-----------------+-----+-----+-----+-----------+


VAL
+-----------------+------------------+-----+-----+-----+-------+
|mean_amt         |std_amt           |p50  |p90  |p99  |max_amt|
+-----------------+------------------+-----+-----+-----+-------+
|29.303233026917116|22.938549105500613|21.78|59.16|105.0|798.59 |
+-----------------+------------------+-----+-----+-----+-------+


TEST
+-----------------+------------------+----+-----+------+---------+
|mean_amt         |std_amt           |p50 |p90  |p99   |max_amt  |
+-----------------+------------------+----+-----+------+---------+
|28.933019441981518|104.39028457909063|21.6|56.34|104.88|335550.94|
+-----------------+------------------+----+-----+------+---------+
```

- **C2.** Robust label clipping (winsorisation at 99.95th percentile)

```
train: pyspark.sql.connect.dataframe.DataFrame = [total_amount: double, trip_distance_km: double ... 11 more fields]
val: pyspark.sql.connect.dataframe.DataFrame = [total_amount: double, trip_distance_km: double ... 11 more fields]
test: pyspark.sql.connect.dataframe.DataFrame = [total_amount: double, trip_distance_km: double ... 11 more fields]
Using label cap (train 99.95th): 134.82
```

Appendix D – Model Performance

- **D1.** RMSE results for baseline, Ridge closed-form, Ridge GD, Huber regression, and Decision Tree fallback

```
=== RMSE summary on TRUE labels (Val | Test) ===
Baseline                     16.948 |  103.391
Model_A_ridge_closed         13.357 |  102.847
Model_B_ridge_gd             18.231 |  103.591
Model_C_huber                29.012 |  105.900
Model_D_tree_fallback        16.652 |  103.267

Best model by validation RMSE: Model_A_ridge_closed
```

- **D2.** Validation vs test comparisons on both TRUE and ROBUST labels

```
(ROBUST RMSE for diagnostics)
Model A: val 12.577 | test 12.611
Model D: val 15.279 | test 15.036
```

Appendix E – Feature Diagnostics

- **E1.** Feature–label correlations (TRUE vs ROBUST)

	feature	1.2 corr_total_amount	1.2 corr_label_robust
1	trip_distance_km_z	0.04	0.92
2	duration_min_z	0.03	0.83
3	pu_te_z	0.02	0.43
4	do_te_z	0.02	0.39
5	passenger_count...	0	0
6	hour_z	0	0
7	month_z	0	0.02
8	dow_z	0	-0.02
9	color_te_z	0	0.04

- **E2.** Standardised coefficients from Ridge regression (Model A)

	feature	coef	abs_coef
1	dist	9.31	9.31
2	dur	4.10	4.10
3	do_te	0.97	0.97
4	color_te	0.88	0.88
5	pc	-0.23	0.23
6	pu_te	0.21	0.21
7	mo	-0.04	0.04
8	hr	0.03	0.03
9	dw	-0.02	0.02

- **E3.** Variance Inflation Factor (VIF) values for multicollinearity check

	feature_z	VIF
1	trip_distance_km_z	3.38
2	duration_min_z	2.71
3	pu_te_z	1.43
4	do_te_z	1.33
5	color_te_z	1.19
6	hour_z	1.02
7	dow_z	1.02
8	passenger_count_z	1.00
9	month_z	1.00